

```
1 n = 3
2 m = 4
3 target = 7
4
5 dp = [[0] * (target + 1) for _ in range(n + 1)]
6 dp[0][0] = 1
7
8 for i in range(1, n + 1):
9     for j in range(1, target + 1):
10         for k in range(1, m + 1):
11             if j >= k:
12                 dp[i][j] += dp[i - 1][j - k]
13
14 print("Number of ways =", dp[n][target])
```

Number of ways = 12

=== Code Execution Successful ===

```
1 n = 5
2 r = 2
3
4 dp = [[0] * (r + 1) for _ in range(n + 1)]
5
6 for i in range(n + 1):
7     dp[i][0] = 1
8
9 for i in range(1, n + 1):
10     for j in range(1, min(i, r) + 1):
11         dp[i][j] = dp[i - 1][j - 1] + dp[i - 1][j]
12
13 print("C(5,2) =", dp[n][r])
```

C(5,2) = 10

=== Code Execution Success

```
1 n = 8
2 r = 3
3
4 dp = [[0] * (r + 1) for _ in range(n + 1)]
5
6 for i in range(n + 1):
7     dp[i][0] = 1
8
9 for i in range(1, n + 1):
10     for j in range(1, min(i, r) + 1):
11         dp[i][j] = dp[i - 1][j - 1] + dp[i - 1][j]
12
13 print("C(8,3) =", dp[n][r])
```

C(8,3) = 56

=== Code Execution Success ===

```
1 n = 10
2 r = 4
3
4 dp = [[0] * (r + 1) for _ in range(n + 1)]
5
6 for i in range(n + 1):
7     dp[i][0] = 1
8
9 for i in range(1, n + 1):
10     for j in range(1, min(i, r) + 1):
11         dp[i][j] = dp[i - 1][j - 1] + dp[i - 1][j]
12
13 print("C(10,4) =", dp[n][r])
```

C(10,4) = 210

=== Code Execution Successful ===

```
1 n = 12
2 r = 5
3
4 dp = [[0] * (r + 1) for _ in range(n + 1)]
5
6 for i in range(n + 1):
7     dp[i][0] = 1
8
9 for i in range(1, n + 1):
10     for j in range(1, min(i, r) + 1):
11         dp[i][j] = dp[i - 1][j - 1] + dp[i - 1][j]
12
13 print("Number of Committees =", dp[n][r])
```

Number of Committees = 792

=== Code Execution Successful ===

```
1 n = 15
2 r = 6
3
4 dp = [[0] * (r + 1) for _ in range(n + 1)]
5
6 for i in range(n + 1):
7     dp[i][0] = 1
8
9 for i in range(1, n + 1):
10     for j in range(1, min(i, r) + 1):
11         dp[i][j] = dp[i - 1][j - 1] + dp[i - 1][j]
12
13 print("C(15,6) =", dp[n][r])
```

C(15,6) = 5005

=== Code Execution Successful ===

```
1 n = 10
2 r = 3
3
4 dp = [[0] * (r + 1) for _ in range(n + 1)]
5
6 for i in range(n + 1):
7     dp[i][0] = 1
8
9 for i in range(1, n + 1):
10     for j in range(1, min(i, r) + 1):
11         dp[i][j] = dp[i - 1][j - 1] + dp[i - 1][j]
12
13 print("Number of Teams =", dp[n][r])
```

Number of Teams = 120

=== Code Execution Successful ===

```
1 n = 12
2 r = 4
3
4 dp = [[0] * (r + 1) for _ in range(n + 1)]
5
6 for i in range(n + 1):
7     dp[i][0] = 1
8
9 for i in range(1, n + 1):
10     for j in range(1, min(i, r) + 1):
11         dp[i][j] = dp[i - 1][j - 1] + dp[i - 1][j]
12
13 print("Number of Committees =", dp[n][r])
```

Number of Committees = 495

=== Code Execution Successful ===


```
1 n = 15
2 r = 5
3
4 dp = [[0] * (r + 1) for _ in range(n + 1)]
5
6 for i in range(n + 1):
7     dp[i][0] = 1
8
9 for i in range(1, n + 1):
10     for j in range(1, min(i, r) + 1):
11         dp[i][j] = dp[i - 1][j - 1] + dp[i - 1][j]
12
13 print("Number of Selections =", dp[n][r])
```

Number of Selections = 3003

=== Code Execution Successful ===

```
1 n = 20
2 r = 6
3
4 dp = [[0] * (r + 1) for _ in range(n + 1)]
5
6 for i in range(n + 1):
7     dp[i][0] = 1
8
9 for i in range(1, n + 1):
10     for j in range(1, min(i, r) + 1):
11         dp[i][j] = dp[i - 1][j - 1] + dp[i - 1][j]
12
13 print("Number of Groups =", dp[n][r])
```

Number of Groups = 38760

=== Code Execution Successful ===

```
1 n = 8
2 r = 3
3
4 dp = [[0] * (r + 1) for _ in range(n + 1)]
5
6 for i in range(n + 1):
7     dp[i][0] = 1
8
9 for i in range(1, n + 1):
10     for j in range(1, min(i, r) + 1):
11         dp[i][j] = dp[i - 1][j - 1] + dp[i - 1][j]
12
13 print("Number of Combinations =", dp[n][r])
```

Number of Combinations = 56

=== Code Execution Successful ===